SRI CHAITANYA EDUCATIONAL INSTITUTIONS

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

NEET GRAND TEST-3

Name :..... Hall Ticket No: | | | | | | | |

INSTRUCTIONS TO CANDIDATES

1. The Model **NEET- 2020** is of **3 Hrs** duration. **Time: 02.00 PM – 05.00 PM**.
2. The question paper for NEET-2020 consists of 180 questions comprising 45 questions in Botany, 45 in Zoology, 45 in Physics and 45 in Chemistry for NEET.
3. All questions are of objective type (Multiple choices only)
4. Each question carries four marks.
5. Negative marking: one mark will be deducted for every wrongly answered question.
6. Total Marks 720.
7. The candidates are prohibited from carrying any paper to the examination hall except **HALL TICKET**.
8. No Calculators, Mini-Cards, Watches with Calculators, Pager, Cell Phone, Slide rules or outer aids to calculation will be allowed in the examination hall.
9. Candidates once admitted will not be allowed to leave the hall till half an hour before the closing of the test.
10. A separate sheet is attached in the middle of this booklet for rough work, you can detach and use it.
11. A detachable answer sheet with 180 question blocks, with 4 circles corresponding to 4 multiple choice for each question will be provided. Use HB Pencil to darken the appropriate circle against the question number provided in the sheet. Answer should be marked only on the answer sheet, but not on the question paper booklet.

1. Nitrogenous waste product of purine metabolism is
   (1) creatinine
   (2) urea
   (3) uric acid
   (4) ammonia
2. Which of the following is not an autoimmune disorder ?
   (1) Rheumatoid arthritis
   (2) Myasthenia gravis
   (3) Myotonic dystrophy
   (4) Addison's disease
3. Statement (SI) : Hypoparathyroidism in humans causes the disorder tetany.
   Statement (SII) : Low calcium levels in blood results in spontaneous generation of action potentials causing muscle contraction
   (1) Both SI & SII are true
   (2) SI is false SII is true
   (3) SI is true, SII is false
   (4) Both SI & SII are false
4. Which one of the following correctly matched with its particular named taxonomic category?
   (1) Housefly- family Diptera
   (2) Tiger- family Carnivora
   (3) Humans- family Primate
   (4) Mango- family Anacardiaceae
5. All are defining characteristic of living organisms except
   (1) Increase in mass from inside
   (2) Reproduction
   (3) Metabolism
   (4) Consciousness
6. Ergot of rye is caused by *Claviceps*. This belongs to
   (1) Ascomycetes
   (2) Basidiomycetes
   (3) Phycomycetes
   (4) Deuteromycetes
7. Identify the correct statement about Eustachian tubes.
   (1) Help in equalizing pressure on two sides of tympanum
   (2) connects middle ear and voice box
   (3) found in all vertebrates
   (4) Connect middle ear cavity to oropharynx
8. Cortisol is responsible for
   (1) causing allergic response
   (2) reducing stress
   (3) glycogenesis and lipogenesis
   (4) causing inflammatory reactions

9. In kidney, the extensions of cortex into medulla are called
   (1) renal columns
   (2) renal pyramids
   (3) renal papillae
   (4) renal calyces
10. All are archaebacteria except
   (1) *Chlorella*
   (2) *Thermococcus*
   (3) *Methanococcus*
   (4) *Methanobacterium*
11. Reserve food in green algae is
   (1) Floridean starch
   (2) Laminarin
   (3) Starch
   (4) Mannitol
12. Read the following statements (A-D) and answer the question which follows them
   A) Sporophyte is predominant phase in pteridophyte
   B) Most pteridophytes are heterosporous
   C) Sexual reproduction in Volvox is oogamous
   D) Sporophyte in *Marchantia* is more elaborate than *Funaria*.

   How many of these statements are correct?
   (1) one
   (2) two
   (3) three
   (4) four
13. Identify the incorrect match from the following pertaining to human heart
   (1) Sinuatrial node – pacemaker
   (2) Eustachian valve – coronary sinus
   (3) Purkinje fibres – specialized muscle fibres
   (4) AV bundle – interventricular septum
14. Which of the following is a dioecious Bryophyte?
   (1) *Marchantia*
   (2) *Cycas*
   (3) Date palm
   (4) all the above
15. Select correct combination regarding respiration
   A) Carbondioxide diffuses 20 times faster than oxygen as it has greater solubility in blood
   B) During normal respiration pressure gradient is created between lungs and atmosphere
   C) Pulmonary volume is altered due to contractions and relaxations of lungs
   D) Contraction of external intercostals muscles and internal intercostal muscles causes normal inspiration and expiration respectively
   (1) A & B
   (2) B & C
   (3) C & D
   (4) A & D

16. How many of these have inferior ovary?

Plum, rose, peach, mustard, china rose, cucumber, brinjal

(1) Two
(2) Three
(3) Four
(4) Five

17. Match the following

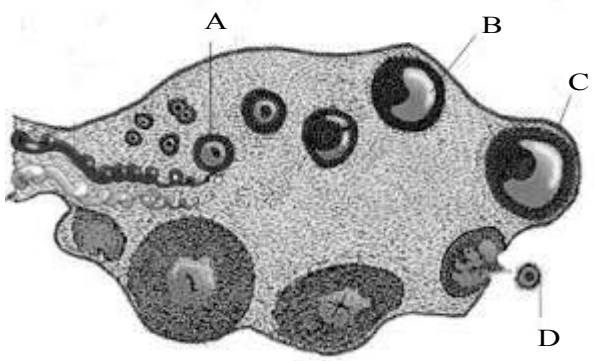
List A	List B
A. Color blindness	I. Inflammatory reactions
B. Night blindness	II. Deficiency of retinol
C. Elephantiasis	III. X-linked recessive genes
D. Ascariasis	IV. Constipation
E. Amoebiasis	V. Internal bleeding

	A	B	C	D	E
(1)	II	III	I	IV	V
(2)	II	III	V	IV	I
(3)	III	II	I	V	IV
(4)	III	I	II	V	IV

18. Testa is edible in

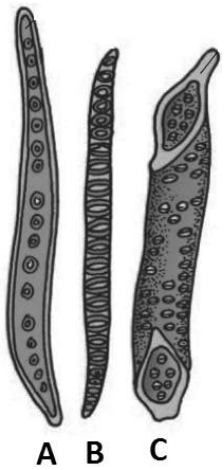
(1) Pomegranate
(2) Tomato
(3) Potato
(4) Banana

19. In the following diagram, identify the mature follicle with secondary oocyte.



(1) A
(2) B
(3) C
(4) D

20. Examine the figure given below and identify correct label



(1) A-pitted tracheid, B-annular tracheid, C pitted vessel
(2) A-Annular tracheid, B-pitted tracheid, C- pitted vessel
(3) A-Vessel, B-tracheid, C-fiber
(4) A-Tracheid, B-fiber, C-vessel

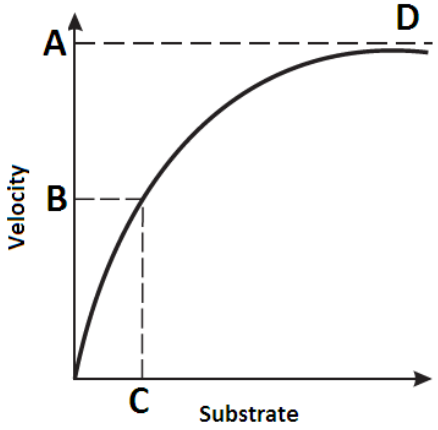
21. Which of the following is not considered a male sex accessory duct ?

(1) Rete testis
(2) Epididymis
(3) Urethra
(4) Vas deferens

22. In assisted reproductive technology procedure fertilization takes place internally in the method

(1) ICSI
(2) IUI
(3) ZIFT
(4) IUT

23. Which of the following is not used as an emergency contraceptive to avoid unwanted pregnancy
    (1) Administration of progestogens
    (2) Administration of progestogen – oestrogen combination
    (3) Intake of "once a week" pill immediately
    (4) Insertion of intra uterine devices
24. D-ribose is
    (1) Keto pentose
    (2) Pentose
    (3) Hexose
    (4) Erythrose
25. Examine the graph showing effect of substrate on enzyme activity



If competitive inhibitor is present then
    (1) A will increase
    (2) B will increase
    (3) C will increase
    (4) all A,B,C will increase
26. Cellular organelles without membranes are
    (1) Centriole, ribosome
    (2) mesosome, chromatophores
    (3) Chloroplast and mitochondria
    (4) ribosome and peroxisome
27. Phenyl ketonuria is not an example of
    (1) pleiotropy
    (2) autosomal recessive trait
    (3) sex linked recessive trait
    (4) inborn error of metabolism
28. Antigenic nature of bacteria is due to
    (1) plasma membrane
    (2) capsule
    (3) nuclear membrane
    (4) DNA
29. Which of the following joints do not permit any movement ?
    (1) Joint between atlas and axis
    (2) Joint between adjacent vertebrae of lumbar region
    (3) Joints between cranial bones
    (4) Joints between carpals
30. Match column-I and Column-II

Column-I	Column-II
a) Zygotene	i) Tetrad become visible
b) Pachytene	ii) Bivalent
c) Diplotene	iii)Terminalisation
d) Diakinesis	iv) chiasmata

    (1) a-I b-ii c-iii d-iv
    (2) a-iv b-iii c-ii d-i
    (3) a-iii b-ii c-i d-iv
    (4) a-ii b-I c-iv d-iii

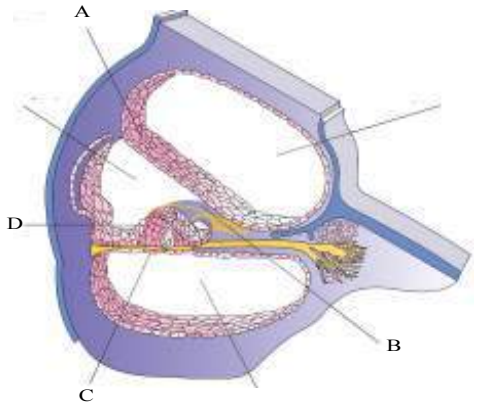
31. Match the following

List – I	List – II
A. Acromion process	I. Fore wings of cockroach
B. Tegmina	II. Scapula bone
C. Erythropoietin	III. Midbrain of humans
D. Cerebral peduncles	IV. JG cells of kidney

	A	B	C	D
(1)	III	II	IV	I
(2)	II	I	IV	III
(3)	II	III	I	IV
(4)	IV	II	I	III

32. Chromosomes attach to spindle fiber by kinetochores in
(1) prophase
(2) metaphase
(3) anaphase
(4) telophase

33. Identify tectorial membrane from the following diagram



(1) A
(2) B
(3) C
(4) D

34. The enzymes that are absent in succus entericus
(1) nucleosidases
(2) nucleotidases
(3) disaccharidases
(4) nucleases

35. Phenomenon associated with root pressure is/are
(1) Bleeding
(2) Guttation
(3) Sap exudation
(4) All of these

36. Factor which is inversely related to transpiration is
(1) absolute humidity
(2) Temperature
(3) both 1 and 2
(4) Light

37. The disease caused by ring worm is known as
(1) dermatitis
(2) dermatomycosis
(3) follicular hyperkeratosis
(4) oriental sores

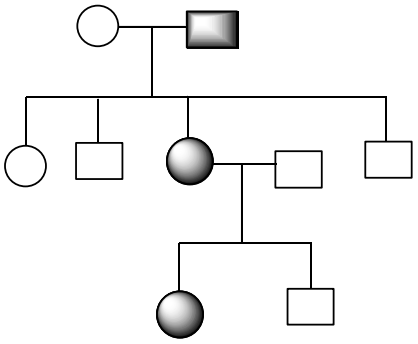
38. Prorennin is secreted by
(1) JG cells
(2) Parietal cells
(3) Paneth cells
(4) Chief cells

39. Choose correct option

Bacteria	Nutrition	N_2 Fixation
(1) Rhizobium	Auto troph	Symbiotic
(2) Frankia	Hetero troph	Symbiotic
(3) Anabaena	Auto troph	Only symbiotic
(4) Azoto bacter	Hetero troph	Symbiotic

40. All are functions of calcium except
    (1) activator of alcohol dehydrogenase
    (2) normal functions of cell membrane
    (3) formation of mitotic spindle
    (4) synthesis of cell wall

41. Study the given pedigree and identify the reason for the appearance of disorder



    (1) X – linked dominant genes
    (2) X – Linked recessive genes
    (3) Y-Linked dominant genes
    (4) Autosomal recessive genes

42. DNA obtained from a single cell is enough to perform DNA finger printing analysis. This is possible mainly due to
    (1) restriction endonucleases
    (2) polymerase chain reaction technique
    (3) r – DNA technology
    (4) Enzyme linked immunosorbent assay

43. In Kranz anatomy, bundle sheath cells
    (1) have thin wall
    (2) have intercellular spaces
    (3) have large number of chloroplast
    (4) lack RUBISCO

44. In C_4 plants CO_2 accepting enzyme
    (1) occur in Bundle sheath cell
    (2) occur in mesophyll cells
    (3) is absent
    (4) Both 1 and 2

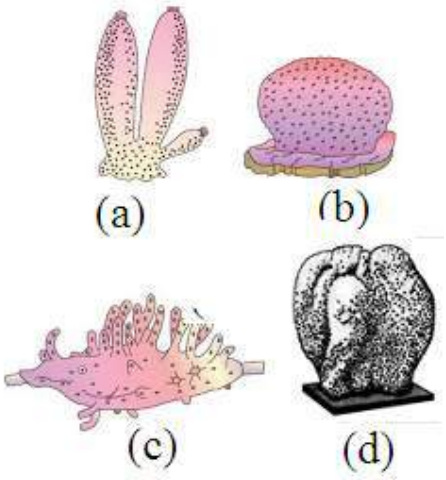
45. The hormone which is not involved in parturition is
    (1) estradiol
    (2) oxytocin
    (3) cortisol
    (4) progesterone

46. Match the following

List – I	List – II
A. *Pristis*	I. Operculum
B. *Pterophyllum*	II. Placoid scales
C. *Pteropus*	III. Pinnae
D. *Octopus*	IV. Stomochord
E. *Saccoglossus*	V. Radula

	A	B	C	D	E
(1)	I	III	II	IV	V
(2)	II	I	III	V	IV
(3)	II	III	IV	V	I
(4)	II	I	IV	II	V

47. How many oxidative decarboxylations occur in mitochondria during aerobic respiration for one pyruvate?
    (1) Two
    (2) Three
    (3) Four
    (4) Six
48. Mobile electron carriers in ETS in mitochondria
    (1) Cytochrome a
    (2) Cytochrome a_3
    (3) Cytochrome c
    (4) Cytochrome c_1
49. Identify the freshwater sponge from the following



    (1) A
    (2) B
    (3) C
    (4) D
50. Select the incorrect statement from the following
    (1) Sea walnuts exhibit bioluminescence
    (2) Sea anemones exhibit metagenesis
    (3) Fertilisation is external in sea urchins
    (4) Excretory system is absent in sea cucumbers
51. Viviparity is totally absent in
    (1) amphibians
    (2) aves
    (3) reptiles
    (4) bony fish
52. Hormone which delay senescence is
    (1) ethylene and ABA
    (2) ABA and GA
    (3) GA and CK
    (4) All the above
53. Auxin can be biossayed by
    (1) *Avena* curvature test
    (2) dwarf pea test
    (3) cotton effect
    (4) triple response test
54. Which of the following is incorrectly matched?
    (1) Rhizome-Ginger
    (2) Leaf buds-water hyacinth
    (3) Bulbil-*Agave*
    (4) conidia-*Penicillium*
55. Identify the incorrect statement from the following
    (1) Pons varolii consists of tracts of nerve fibres that connect different parts of brain.
    (2) Cerebrum integrates information from semicircular canals of the ear and auditory system.
    (3) Limbic system is concerned with olfaction, motivation and autonomic responses.
    (4) Cerebellum is not a part of brain stem

56. How many sets of chromosomes are present in meiocyte of Onion?
(1) 16
(2) 8
(3) 32
(4) 2

57. Select the correct option about different stages of cross bridge cycle.
A. Formation of cross bridge
B. Breaking of cross bridge
C. Binding of ATP to myosin head
D. Sliding or rotation of cross bridge
E. ATP hydrolysis
(1) A – C – B – D – E
(2) A – D – E – B – C
(3) E – A – D – C – B
(4) A – E – D – B – C

58. Insect pollinated plant among the following are
(1) *Yucca, Salvia, Ficus*
(2) *Santalum, Michelia*
(3) *Arisaema Adansonia*
(4) *Anthocephalus, Bombax, Callistemon*

59. National forest policy of India recommended percentage of forest cover for plains and hills in
(1) 1980
(2) 1975
(3) 1981
(4) 1988

60. Ploidy level of archesporium cell in nucellus is
(1) haploid
(2) diploid
(3) triploid
(4) Tetraploid

61. Gause principle is related to
(1) competition between closely related species for limited natural resources
(2) resource partitioning by different species
(3) interference competition under unlimited resources
(4) intraspecific competition for food, space and mating

62. How many types of gametes are possible from a diploid organism having genotype AaBbCc ?
(1) 2
(2) 4
(3) 8
(4) 16

63. Study the following statements and how many are related to *Australopithecines*
A. They lived in east African grasslands
B. They were hairy and walked like gorillas and chimpanzees
C. They hunted with stone weapons
D. They probably were meat eating
E. They were called first human beings as they walked upright
(1) Four
(2) Two
(3) Three
(4) Only one

64. How many alleles are present in 2n cell for ABO blood group?
(1) Six
(2) Four
(3) Three
(4) Two

65. Which of the following is considered as the functional aspect of an ecosystem ?
(1) Species composition
(2) Species diversity
(3) Stratificaiton
(4) Decomposition

66. The ability of a gene to have multiple phenotypic effect is known as
(1) Polygenic inheritance
(2) Pleiotropy
(3) Incomplete dominance
(4) Polytypic effect

67. Muscles responsible for blood circulation in cockroach are
(1) dorso ventral muscles
(2) cardiac muscles
(3) alary muscles
(4) papillary muscles

68. In human body, the tissue found in dermis of skin is
(1) stratified epithelial tissue
(2) adipose tissue
(3) irregular dense fibrous tissue
(4) areolar tissue

69. Splicing occur in cells of
(1) *Mycoplasma*
(2) *Nostoc*
(3) *Chlamydamonas*
(4) *Anabaena*

70. Which of the following bond is not present in DNA?
(1) H-bonds
(2) Phosphodiester bond
(3) glycosidic bond
(4) peptide bond

71. Identify the correct statement pertaining to frog
(1) Conus arteriosus is present on dorsal side of heart
(2) Sinus venosus is present on ventral side of heart
(3) Bidder's canal is present in the kidney of male frogs
(4) In female frogs, a pair of oviducts arising from ovaries open separately to outside

72. In Meghalaya, large number of rare and threatened plants are preserved in
(1) Western Ghats
(2) Aravalli hills
(3) Bastar & Chanda areas
(4) Khasi & Jaintia hills

73. Which of the following enhances or induces fusion of protoplasts?
(1) potassium chloride
(2) sodium chloride
(3) 1AA, Kinetin
(4) Polyethylene glycol

74. Pusa A-4 variety is resistant to shoot and fruit borer. This is variety of
(1) *Brassica*
(2) Flats bean
(3) Okra (Bhindi)
(4) Chilli

75. The biological product $\alpha-1-$ antitrypsin, synthesized by using transgenic animals, helps in treatment of the disease
(1) cystic fibrosis
(2) alkaptonuria
(3) phenyl ketonuria
(4) emphysema

76. In Australia, which of the following pairs indicate convergent evolution ?
(1) Wombat and Numbat
(2) Sugar glider and Flying phalanger
(3) Lemur and spotted cuscus
(4) Thylacine and Tasmanian tiger cat

77. Source of cyclosporin A is
(1) Virus
(2) Bacteria
(3) Fungi
(4) Algae

78. After sewage treatment which of the following is generally released into natural water body?
(1) effluent from primary treatment plant
(2) effluent from secondary treatment plant
(3) activated sludge
(4) floc

79. Match the following

List I	List II
A. Scrubber	I. Plastic wastes
B. DDT	II. Sulphurdioxide
C. Nitrates & Phosphates	III. Biomagnification
D. Polyblend	IV. Plant nutrients

	A	B	C	D
(1)	I	III	II	IV
(2)	II	III	IV	I
(3)	II	I	III	IV
(4)	III	I	II	IV

80. In animal husbandry, size of herd can be increased in a short period of time by
(1) Artificial insemination
(2) Inter specific hybridization
(3) Multiple ovulation embryo transfer technology
(4) In-vitro fertilization embryo transfer technology

81. Integrated pest management is _____method of pest management
(1) Chemical
(2) Cultural and biological
(3) Physical and mechanical
(4) all the above

82. An enzyme catalysing removal of nucleotide from ends of DNA is
(1) exopeptidase
(2) Hind II
(3) Hind III
(4) exonuclease

83. How many of the following animals may occupy more than one trophic level in same ecosystem ?
A) *Passer* B) *Psittacula*
C) *Periplaneta* D) *Oryctolagus*
E) *Antelope* F) *Homo sapiens*
(1) Three
(2) Four
(3) Two
(4) Five

84. The transfer of genetic material from one bacterium to another through the medium of vector like virus is termed as
(1) Virulence
(2) Conjugation
(3) Transduction
(4) Translation

85. Identify the euryhaline animal from the following
(1) *Scoliodon*
(2) *Salmon*
(3) *Sardine*
(4) *Salpa*

86. DNA can be visualized on agarose gel by staining with _____ and exposing in UV rays.
(1) ethidium bromide
(2) safranine
(3) fast green
(4) janus green

87. Choose the incorrect match from the following
(1) Pro insulin – C peptide
(2) Retrovirus – gene therapy
(3) Pomfrets – edible oysters
(4) *Apis indica* – haplodiploidy

88. i) Examine the validity of genetic modification of organism research
ii) inspect the safety of introducing GMO for public services
These are objective of
(1) GEAC
(2) ACMR
(3) IARI
(4) MIPG

89. Statement (SI) : Excessive dosage of cocaine causes hallucinations
Statement(SII) : Cocaine acts as a depressant and induces sleep
(1) Both SI & SII are true
(2) SI is false SII is true
(3) SI is true, SII is false
(4) Both SI & SII are false

90. Example of a sedimentary cycle with reservoir located in earth's crust are
(1) Nitrogen and sulphur
(2) Nitrogen and carbon
(3) Carbon and phosphorus
(4) Sulphur and phosphorus

91. Which of the following has its unit Newton-Second
(1) Energy
(2) Torque
(3) Momentum
(4) Power

92. Two moles of ideal helium gas is in a rubber balloon at 30^0C. The balloon is filled expandable and can be assumed to require no energy in its expansion. The temperature of the gas in the balloon is changed to 35^0C. The amount of heat required in raising the temperature is nearly (R=8.3 J/mole-k)
(1) 62 J
(2) 104 J
(3) 124 J
(4) 208 J

93. One end of a horizontal thick copper wire of length 2l and radius 2r is welded to the end of another thin horizontal copper wire of length ℓ and radius r. When the arrangement is stretched by applying forces at two ends the ratio of elongation in thin wire to that in the thick wire is

(1) 0.25

(2) 0.50

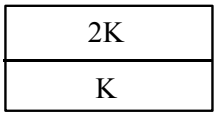
(3) 2.00

(4) 4.00

94. Two rectangular blocks of same dimensions and conductivities K and 2K are arranged as shown. The temperature difference between ends are same. If it takes 9 sec to transport certain amount of heat in the first condition, the time taken by same amount of heat to transport in the second combination



(1)



(2)

(1) 2 sec

(2) 3 sec

(3) 4.5 sec

(4) 6 sec

95. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\bar{E} = E\hat{i}$ and magnetic field $\bar{B} = B\hat{i}$ with velocity $\bar{\upsilon} = \upsilon_0\hat{j}$. The speed of the particle will become $2\upsilon_0$ after a time.

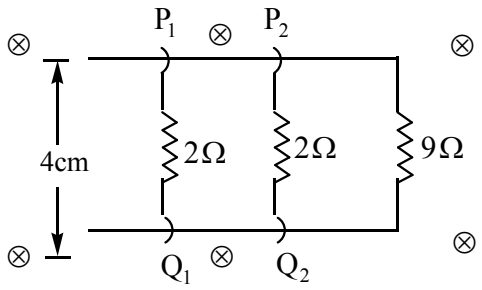
(1) $t = \frac{2m\upsilon_0}{qE}$

(2) $t = \frac{2Bq}{m\upsilon_0}$

(3) $t = \frac{\sqrt{3}Bq}{m\upsilon_0}$

(4) $t = \frac{\sqrt{3}m\upsilon_0}{qE}$

96. In the diagram P_1Q_1 and P_2Q_2 are made to slide on the rails with same speed of 5 cm/s towards left. In this region a magnetic field of 1T exists. The current in 9Ω resistor is



(1) zero

(2) 0.2 mA

(3) 0.1 mA

(4) 0.4 mA

97. 'n' molecules of a gas are enclosed in a cubical vessel of edge length *l*, then pressure exerted by the gas if each molecule has mass m and rms speed is u, in

(1) $\frac{mn\upsilon^2}{l^3}$

(2) $\frac{m^2n\upsilon}{2l^3}$

(3) $\frac{mn\upsilon^2}{3l^3}$

(4) $\frac{m^2n\upsilon}{2l}$

98. Two bodies each of mass 'M' are separated by a distance 2*l*. A particle of mass m is projected from the midpoint of the line joining their centres perpendicular to the line. The minimum initial speed of the mass m to escape from the gravitational field of the two bodies is

(1) $4\sqrt{\frac{GM}{l}}$

(2) $2\sqrt{\frac{GM}{l}}$

(3) $4\sqrt{\frac{2GM}{l}}$

(4) $\sqrt{\frac{GM}{l}}$

99. The rms velocity of the molecules in a sample of helium is 5/7 in that of the molecules in a sample of hydrogen. If the temperature of hydrogen sample is 0^0C. The temperature of the helium sample is nearly

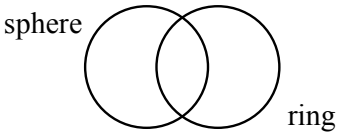
(1) 0^0C

(2) 0 K

(3) 273^0C

(4) 60^0C

100. A charge q is distributed uniformly on a ring of radius r. A sphere of equal radius is constructed with its centre at the periphery of the ring. The electric flux through the surface of the sphere as



(1) $\frac{q}{\epsilon_0}$

(2) $\frac{2q}{\epsilon_0}$

(3) $\frac{q}{2\epsilon_0}$

(4) $\frac{q}{3\epsilon_0}$

101. A charge q is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium if q is equal to

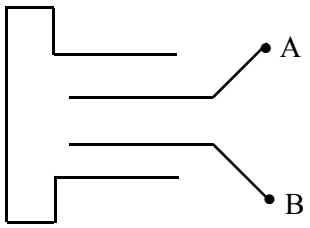
(1) $-\frac{Q}{2}$

(2) $-\frac{Q}{4}$

(3) $+\frac{Q}{4}$

(4) $+\frac{Q}{2}$

102. Find the effective capacity between A and B.



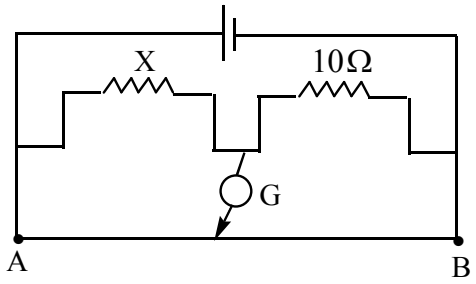
(1) $\frac{3}{2}$C

(2) 3 C

(3) $\frac{3}{5}$C

(4) $\frac{2}{3}$C

103. A meter bridge is set up as shown to determine an unknown resistance X using a standard resistance 10Ω. The galvanometer shows null point when tapping jockey is at 52cm mark. The end corrections are 1cm as 2cm respectively for the ends A and B. The determine value of X is



(1) 10.6Ω
(2) 10.8Ω
(3) 10.10Ω
(4) 11.2Ω

104. A satellite of mass m is orbiting the earth of radius R at a height h from its surface. The total energy of the satellite in terms of g_o, the value of acceleration due to gravity at the earth's surface, is

(1) $\frac{mg_0R^2}{2(R+h)}$

(2) $\frac{-mg_0R^2}{2(R+h)}$

(3) $\frac{2mg_0R^2}{R+h}$

(4) $\frac{-2mg_0R^2}{(R+h)}$

105. The potential is represented by $v = 4x^2$ volt. Calculate magnitude of the electric field at (1m, 0m, 2m)

(1) 8 v/m
(2) 16 v/m
(3) 0 v/m
(4) 4 v/m

106. Magnetic susceptibility is negative for

(1) Ferro substances
(2) Para substances
(3) Dia substances
(4) Para and Ferro substances

107. The efficiency of a carnot heat engine is 1/10. It is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at low temperature is

(1) 10 J
(2) 30 J
(3) 60 J
(4) 90 J

108. Two soap bubbles of radii a and b (a < b) are formed one inside the other. The radius of another soap bubble which has same excess pressure as that of difference in pressure between inside smaller bubble and out side the larger bubble is

(1) $\frac{a+b}{a+b}$

(2) a + b

(3) $\frac{ab}{a+b}$

(4) $\frac{a^2b}{(a+b)^2}$

109. A vessel is filled with two immisable liquids of densities d and 2d each of height H/2 where H is the height of the vessel. A hole in made to the wall of the vessel at a height h from the bottom (h < H/2). The velocity of efflux of the liquid is

(1) $\sqrt{\frac{(H-h)g}{2}}$

(2) $\sqrt{\frac{(3H-4h)g}{2}}$

(3) $\sqrt{\frac{(2H-3h)g}{2}}$

(4) $\sqrt{2gh}$

110. Compare the current sensitivities of two moving coil galvanometer with the following parameter.

(I) $N_1 = 30$, $A_1 = 1.5\times10^{-3}\,m^2$, $B_1 = 0.25T$; $R_1 = 20\,\Omega$

(II) $N_2 = 35$, $A_2 = 2\times10^{-3}\,m^2$, $B_2 = 0.25T$; $R_2 = 30\,\Omega$

The torsional constants for the suspension fibres in both are same.

(1) 3 : 9

(2) 9 : 14

(3) 14 : 9

(4) 7 : 3

111. The rms voltages V_L, V_C and V_R are across inductor L, capacitor C and resistance R in LCR series circuit. Their ratio is 1 : 2 : 3. If the rms voltage of AC source is 100 V, then V_R is nearly equal to

(1) 40 V

(2) 60 V

(3) 90 V

(4) 100 V

112. The internal resistance of a cell of 2.1 V which gives a current of 0.2 A through a resistance of $10\,\Omega$ is

(1) $0.8\,\Omega$

(2) $1.0\,\Omega$

(3) $0.2\,\Omega$

(4) $0.5\,\Omega$

113. The position of a particle is given by $x = (t-2)^2$ where x is in meres and t in seconds. The distance covered by the particle in first 4 seconds is

(1) 4 m

(2) 8 m

(3) 12 m

(4) 16 m

114. A projectile is moving at $60 ms^{-1}$ at its highest point, where it breaks into two equal parts due to an internal explosion. One part moves vertically up at $50\ ms^{-1}$ w.r.t. ground. The other part will move at

(1) $110\ ms^{-1}$

(2) $120\ ms^{-1}$

(3) $130\ ms^{-1}$

(4) $10\sqrt{61}\ ms^{-1}$

115. In a P-n junction diode, the thickness of depletion layer is $2 \times 10^{-6} m$ and barrier potential is 0.3v. The Intensity of the electric field at the junction is ---

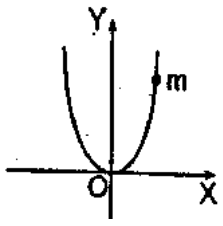
(1) $0.6\times10^{-6}\,v/m$ from n to p side

(2) $0.6\times10^{-6}\,v/m$ from p to n side

(3) $1.5\times10^{5}\,v/m$ from n to p side

(4) $1.5\times10^{5}\,v/m$ from p to n side

116. A bead of mass m is located on a parabolic wire with its axis vertical and vertex directed downward as in figure and whose equation is $x^2 = ay$. If the coefficient of friction is μ, the highest distance above the x-axis at which the particle will be in equilibrium is



(1) μa

(2) $\mu^2 a$

(3) $\frac{1}{4}\mu^2 a$

(4) $\frac{1}{2}\mu a$

117. For a transistor to work as an amplifier
(1) Its emitter and collector junctions are in forward bias
(2) It's emitter junction is in reverse bias and collector junction is in for ward bias
(3) It's emitter junction is in forward bias and collector junction is in reverse bias
(4) The transistor must have break down region

118. A Neutron's moving with a certain kinetic energy collide head on with an atom of mass number 'A'. Find the fraction of kinetic energy retained by it
(1) $\frac{A-1}{A+1}$
(2) $\left(\frac{A-1}{A+1}\right)^3$
(3) $\left(\frac{A-1}{A+1}\right)^2$
(4) $\frac{4A}{(A+1)^2}$

119. In an electromagnetic wave, the $\vec{E}$ and $\vec{B}$ vectors at any instant of time have
(1) same phase
(2) same amplitude
(3) phase difference π
(4) phase difference $\frac{\pi}{2}$

120. A uniform chain of length L and mass M is lying on a smooth table and one third of its length is hanging vertically down over the edge of the table. If g is acceleration due to gravity, the work required to pull the hanging part on to the table is
(1) MgL
(2) $MgL/3$
(3) $MgL/9$
(4) $MgL/18$

121. A particle is projected at time t=0 from a point 'O' with a speed 'u' at an angle 'θ' to horizontal. The torque of gravitational force on projectile about the origin at time 't' is
(1) $(mgu\cot\theta)t$
(2) $(mgu\cos\theta)t$
(3) $(mgu\sin\theta)t$
(4) $(mguTan\theta)t$

122. A disc of radius r and mass m rotating with an angular velocity w is placed on a rough horizontal surface. The initial velocity of the centre of the disc is zero. What will be the velocity of centre of disc when it ceases to slip?
(1) $\frac{rw}{4}$
(2) $\frac{rw}{3}$
(3) $\frac{rw}{2}$
(4) rw

123. A disc at rest at the top of the inclined plane of height 'h' rolls down without slipping and acquires a velocity 'v' on reaching the bottom. If the same disc slides down a smooth incline and acquires the same velocity on reaching the bottom the height of smooth incline is
(1) h/3
(2) h/2
(3) $\frac{2h}{3}$
(4) h

124. The damping force on an oscillator is directly proportional to the velocity. The unit of the constant of proportionality is
(1) kgs^{-1}
(2) kgs
(3) $kgms^{-1}$
(4) $kgms^{-2}$

125. A simple pendulum has time period 'T_1'. The point of suspension is now moved upwards according to the relation $y = kt^2$, $(k = 1m/\sec^2)$ where y is the vertical displacement. The time period now becomes 'T_2' then find the ratio of $\frac{T_1^2}{T_2^2}$ $(g = 10m/\sec^2)$
(1) $\frac{5}{6}$
(2) $\frac{6}{5}$
(3) $\frac{1}{3}$
(4) $\frac{3}{4}$

126. A boy is walking away from a wall towards an observer at a speed of 1 ms^{-1} and blows a whistle whose frequency is 680Hz. The number of beats heard by the observer per second is (Velocity of sound in air =340 ms^{-1})
(1) 4
(2) 8
(3) 2
(4) 10

127. A string is rigidly tied at two ends and its equation of vibration is given by $y = \cos 2\pi t \sin 2\pi x$. The minimum length of the string is
(1) 1 m
(2) 0.5 m
(3) 5 m
(4) $2\pi m$

128. In dispersion without deviation of white light by combination two thin prisms
(1) the emergent rays of all the colours are parallel to the incident ray
(2) yellow coloured ray is parallel to the incident ray
(3) only red coloured ray is parallel to the incident ray
(4) all the rays are parallel, but not parallel to the incident ray

129. A transparent cube $(\mu = 1.5)$ of 15 cm edge contains a small air bubble. Its apparent depth when viewed through one face is 6 cm then when it viewed from the opposite face the respective apparent depth is
(1) 9 cm
(2) 4 cm
(3) 6 cm
(4) 8 cm

130. In youg's double slit experiment how many maximas can be obtained on a screen (including the central maximum) on both sides of the central fringes if $\lambda = 2000\ \text{Å}$ and $d = 7000\ \text{Å}$
(1) 12
(2) 7
(3) 18
(4) 4

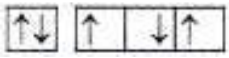
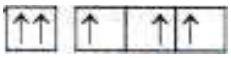
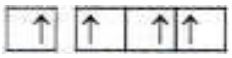
131. If an unpolarised light is converted into plane polarized light, its intensity becomes
(1) Half
(2) One-fourth
(3) One-third
(4) One-eighth

132. As an electron makes a transition from an exited state to the ground state of hydrogen-like atom/ion:
(1) Kinetic energy, potential energy and total energy decrease
(2) Kinetic energy decreases, potential energy increases and total energy remains same
(3) Kinetic energy and total energy decrease but potential energy increases
(4) Its Kinetic energy increases but potential energy and total energy decrease

133. The stopping potential V for photoelectric emission from a metal surface is plotted along Y-axis and frequency ν of incident light along X-axis. A straight line is obtained as shown. Planck's constant is given by
(1) Slope of the line
(2) Product of slope on the line and charge on the electron
(3) Product of intercept along Y-axis and mass of the electron
(4) Product of Slope and mass of electron

134. Half life of a substance a is 20 minutes what is the time between 33% decay and 67% decay?
(1) 40 minutes
(2) 20 minutes
(3) 30 minutes
(4) 25 minutes

135. The operation of a nuclear reactor is said to be critical , if the multiplication factor (k) has a value
(1) 1
(2) 1.5
(3) 2.1
(4) 2.5

136. Which of the following violates the pauli exclusion principle?
(1) 
(2) 
(3) 
(4) 

137. Regarding Bohr's model of atom, true statements are
a) velocity of electron $\propto \frac{1}{n}$
b) energy of electron $\propto -\frac{1}{n^2}$
c) radius of orbit $\propto n^2$
(1) a,b,c
(2) a,b only
(3) b,c only
(4) a,c only

138. Among the following compounds, which of the following has the maximum number of sp-hybridised carbon atoms?

(1) $(CN)_2$

(2) $CH_2 = C = CH - CN$

(3) $HC \equiv C - CH_2 - CH = C = C = CH_2$

(4) $HC \equiv C - CN$

139. which among the following on α-decay will give 2 noble gas nuclides?

(1) Radon

(2) Ceasium

(3) Radium

(4) Thorium

140. Under similar conditions which of the following gases will diffuse four times as quickly as oxygen ?

(1) He

(2) H_2

(3) N_2

(4) D_2

141. Volume strength of perhydrol is:

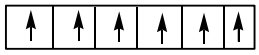
(1) 100 vol

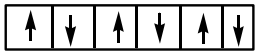
(2) 10 vol

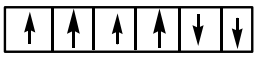
(3) 65 vol

(4) 75vol

142. Column-I Column-II

I)  a) Fe

II)  b) MnO

III)  c) Fe_3O_4

Correct match is

(1) I-b, II-a, III-c

(2) I-c, II-b, III-a

(3) I-a, II-b, III-c

(4) I-c, II-a, III-b

143. HX is a weak mono basic acid whose degree of ionisation is 0.3. Then the freezing point of 0.2 m aqueous solution is [K_f of water = 1.85 $k.m^{-1}$]

(1) – 4.8^0C

(2) – 0.48^0C

(3) – 0.36^0C

(4) – 0.64^0C

144. $C_3H_6 + H_2 \longrightarrow C_3H_8$ ΔH = -224KJ

$C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$; ΔH=-2027KJ

$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$ ΔH = -282KJ,

Based on above data , calculate molar heat of combustion of propene.

(1) -1020 KJ

(2) -1969 KJ

(3) -2020KJ

(4) 765.9 KJ

145. pH of buffer solution containing 20mL of 0.2M CH_3COOH and 100mL 0.4M CH_3COONa is(pK_a = 4.8)

(1) 5.8

(2) 4.8

(3) 9.4

(4) 7.0

146. Statement-I Specific conductance of 0.1MHCl is greater than that of 0.1M CH_3COOH

Statement-II : Increase in molar conductance with dilution is more in 0.1M CH_3COOH than in 0.1M HCl

(1) Both I and II are false

(2) Both I and II are true

(3) I is true, II is false

(4) I is false, II is true

147. During electrolysis, 2A current is passed through an electrolytic solution for 965 sec. The number of moles of electrons passed will be
(1) 0.02
(2) 0.01
(3) 200
(4) 0.037

148. In petrochemical industries, alcohols are directly converted to gasoline by passing over heated:
(1) Pt
(2) Ni
(3) ZSM-5
(4) Fe

149. For a graph plotted between $\log\frac{a}{a-x}$ (y-axis) versus time for a first order reaction. Slope will be
(1) $-k$
(2) $\frac{-k}{2.303}$
(3) $\frac{k}{2.303}$
(4) k

150. Fe_2O_3 is reduced to spongy iron near the top of blast furnace by
(1) CO
(2) CO_2
(3) C
(4) $CaCO_3$

151. which among the following is incorrect statement?
(1) Covalency of nitrogen is 3 in CH_3CN
(2) Central atom in ClF_3 is sp^2 hybridised
(3) NaCN contains both ionic and covalent bonds
(4) Bond order in O_2^{+1} is 2.5

152. Incorrect statement about phosphine is
(1) It is used in smoke screens and Holme's signals.
(2) It is less soluble in water than NH_3
(3) It reacts with $CuSO_4$ to give Cu_3P_2
(4) PH_3 containing P_4 and P_2H_4 is highly stable

153. Which of the following imparts brick red colour to flame?
(1) $CaCl_2$
(2) $BaBr_2$
(3) $CsCl$
(4) $MgSO_4$

154. Pick out the correct statements
(A) In the estimation of ozone, the solution of ozone is first treated with excess KI at a P^H = 9.2 in first borate buffer
(B) Liquid SO_2 is a very good non aqueous solvent for both ionic and covalent compounds
(C) Moist chlorine is a permanent bleaching agent
(D) HOF is a monobasic oxyacid
(1) A,B only
(2) A,B,C only
(3) B,D only
(4) A,B,D only

155. Highest oxidation state exhibited by an actinoid is
(1) +5
(2) +3
(3) +6
(4) +7

156. In modern periodic table, lanthanides are placed in

(1) 3rd group and 7th period

(2) 3rd group and 6th period

(3) 4th group and 7th period

(4) 3rd group and 5th period

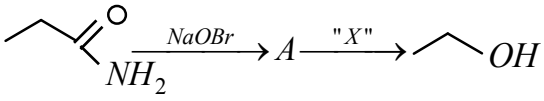
157. Which of the following complex ion will have the maximum number of unpaired electrons?

(1) $[CrCl_6]^{3-}$

(2) $[Cr(H_2O)_6]^{3+}$

(3) $[Cr(CN)_6]^{3-}$

(4) All have same number of unpaired electrons

158. 

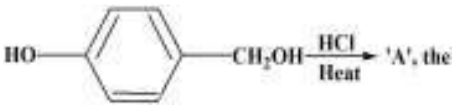
X is

(1) aq.KOH

(2) dil.H_2SO_4

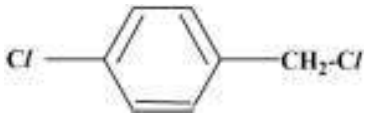
(3) moist Ag_2O

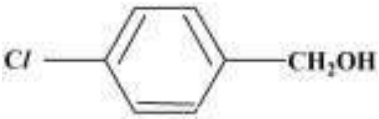
(4) HNO_2

159. 

Structure of compound 'A' is

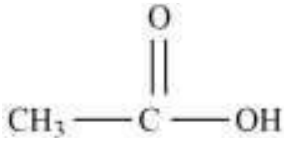
(1) 

(2) 

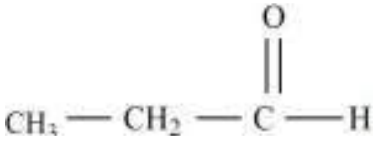
(3) 

(4) No reaction takes place

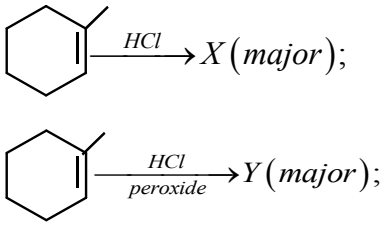
160. Which of the following gives a yellow precipitate with I_2/NaOH:

(1) 

(2) $CH_3 - CH_2OH$

(3) 

(4) All of the above

161. 

X and Y are

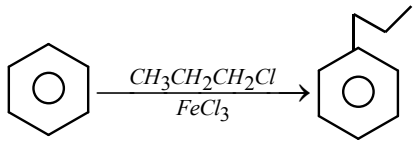
(1) Same compounds

(2) Homologues

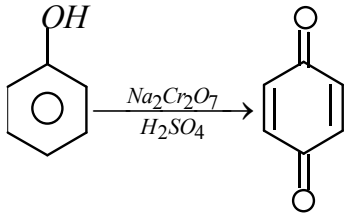
(3) Chain isomers

(4) Positional isomers

162. Which of the following is not a correctly represented reaction

(1) 

(2) $CH_3 - CO - CH_2 - CH_3 \xrightarrow[H_2SO_4]{KMnO_4} 2CH_3COOH$

(3) 

(4) $(CH_3)_3CH \xrightarrow{KMnO_4} (CH_3)_3COH$

163. Which of the following vitamin is anti – oxidant?

(1) Vitamin – A

(2) Vitamin – K

(3) Vitamin – D

(4) Vitamin – E

164. Polyethene, PVC, Teflon and neoprene are all
(1) Condensation homopolymers
(2) Addition homopolymers
(3) Biodegradable polymers
(4) Addition copolymers

165. All the following are Bactericidal antibiotics except
(1) penicillin
(2) Tetracyline
(3) Oflaxacin
(4) Amino glycoside

166. The number of molecules in 22.4 cm^3 of nitrogen gas at STP is
(1) 6.022×10^{20}
(2) 6.022×10^{23}
(3) 22.4×10^{20}
(4) 22.4×10^{23}

167. $A^-_{(g)} \xrightarrow{\Delta H=+360KJ} A(g)$

$A(g) \xrightarrow{\Delta H=+1260KJ} A^+(g)$

Electron gain enthalpy of "A" is
(1) -1260KJ
(2) -1620KJ
(3) -360KJ
(4) -900KJ

168. The observed dipole moment of a compound 'AB' is 2.4Debye and A-B bond length is $1A^0$, then the percent ionic character in AB is:
(1) 25 %
(2) 50%
(3) 75 %
(4) 66.6%

169. Bond enthalpies of N_2 and H_2 are 945KJ/mole and 436 KJ/mole respectively, N-H bond enthalpy is (given $N_2+3H_2\rightarrow 2NH_3+94KJ$)
(1) 781KJ
(2) 195KJ
(3) 391KJ
(4) 585 KJ

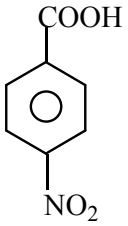
170. A complex with the formula $\left[Cr(H_2O)_4\,Br(ONO)\right]Cl_2$ can exhibit
A) Geometrical isomerism
B) Linkage isomerism
C) Hydrate isomerism
D) Ionization isomerism
(1) A,B,C,D
(2) A,B only
(3) A,B,C only
(4) B,C,D only

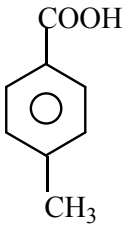
171. For the reaction
$AB(g) \rightleftharpoons A(g)+B(g)$, AB is 33% dissociated at total pressure of P, then
(1) $p = Kp$
(2) $p = 4Kp$
(3) $p = 3Kp$
(4) $p = 8Kp$

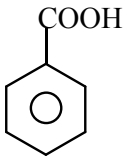
172. $(CH_3)_2Cd \xrightarrow{C_2H_5-\overset{O}{\overset{\|}{C}}-Cl} A \xrightarrow{NaBH_4} B \xrightarrow{Cu/573K} C.$

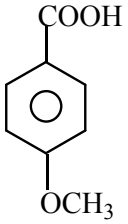
"C" cannot react with
(1) 2,4-DNP
(2) $Na_2CO_3+I_2$
(3) $\left[Ag(NH_3)_2\right]^+$
(4) $NaHSO_3$

173. P^{ka} is highest for

(1) 

(2) 

(3) 

(4) 

174. Arrange H_2O, $HC \equiv CH, NH_3, CH_3CH_3$ in order of acidity

(1) $H_2O > HC \equiv CH > NH_3 > CH_3CH_3$

(2) $HC \equiv CH > H_2O > NH_3 > CH_3 - CH_3$

(3) $CH_3CH_3 > HC \equiv CH > NH_3 > H_2O$

(4) $H_2O > NH_3 > HC \equiv CH > CH_3 - CH_3$

175. For an octahedral complex with $\Delta_0 < P$, the value of CFSE for d^5 configuration is

(1) $-2\Delta_0$

(2) $-3\Delta_0$

(3) $-1.6\Delta_0$

(4) Zero

176. Which of the following is not a greenhouse gas?

(1) CO

(2) O_3

(3) CH_4

(4) H_2O vapour

177. Geometrical isomerism cannot be exhibited by

(1) Crotonaldehyde

(2) Acetylene dichloride

(3) Acetaldoxime

(4) Acetone hydrazone

178. Boric acid is an acid because its molecule

(1) Contains replaceable H atom

(2) Gives up a proton

(3) Accepts OH^- from water leading to release of proton from water.

(4) Combines with proton from water molecule.

179. C-O cleavage of CH_3COOH occurs when it reacts with

(1) CH_3MgBr

(2) $NaHCO_3$

(3) KOH

(4) NH_3 / Δ

180. Which of the following is a high – spin complex?

(1) $\left[Co\left(NH_3\right)_6\right]^{3+}$

(2) $\left[Fe\left(CN\right)_6\right]^{4-}$

(3) $\left[CoF_6\right]^{4-}$

(4) $\left[Zn\left(NH_3\right)_6\right]^{2+}$

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI

SR ELITE, AIIMS S60, NEET MPL & LTC-IC Date : 22.04.20

NEET GRAND TEST – 03 KEY

BOT:- 4, 5, 6, 10, 11, 12, 14, 16, 18, 20, 24, 25, 26, 28, 30, 32, 35, 36, 39, 40, 43, 44, 47, 48, 52, 53, 54, 56, 58, 60, 62, 64, 66, 69, 70, 73, 74, 77, 78, 81, 82, 84, 86, 88, 90

ZOO:- 1, 2, 3, 7, 8, 9, 13, 15, 17, 19, 21, 22, 23, 27, 29, 31, 33, 34, 37, 38, 41, 42, 45, 46, 49, 50, 51, 55, 57, 59, 61, 63, 65, 67, 68, 71, 72, 75, 76, 79, 80, 83, 85, 87, 89

BIOLOGY

1) 3	2) 3	3) 1	4) 4	5) 2	6) 1	7) 1	8) 2	9) 1	10) 1
11) 3	12) 2	13) 2	14) 1	15) 1	16) 1	17) 3	18) 1	19) 3	20) 1
21) 3	22) 2	23) 3	24) 2	25) 3	26) 1	27) 3	28) 2	29) 3	30) 4
31) 2	32) 2	33) 2	34) 4	35) 4	36) 1	37) 2	38) 4	39) 2	40) 1
41) 4	42) 2	43) 3	44) 4	45) 4	46) 2	47) 2	48) 3	49) 3	50) 2
51) 2	52) 3	53) 1	54) 2	55) 2	56) 4	57) 3	58) 1	59) 4	60) 2
61) 1	62) 3	63) 2	64) 4	65) 4	66) 2	67) 3	68) 3	69) 3	70) 4
71) 3	72) 4	73) 4	74) 3	75) 4	76) 3	77) 3	78) 2	79) 2	80) 3
81) 4	82) 4	83) 1	84) 3	85) 2	86) 1	87) 3	88) 1	89) 3	90) 4

PHYSICS

91) 3	92) 4	93) 3	94) 1	95) 4	96) 2	97) 3	98) 2	99) 1	100) 4
101) 2	102) 1	103) 1	104) 2	105) 1	106) 3	107) 4	108) 3	109) 2	110) 2
111) 3	112) 4	113) 2	114) 3	115) 3	116) 3	117) 3	118) 3	119) 1	120) 4
121) 2	122) 2	123) 3	124) 1	125) 2	126) 1	127) 2	128) 2	129) 2	130) 2
131) 1	132) 4	133) 2	134) 2	135) 1					

CHEMISTRY

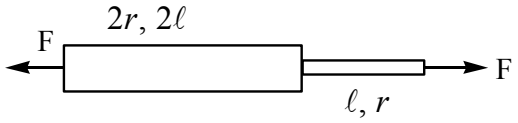
136) 3	137) 1	138) 3	139) 3	140) 2	141) 1	142) 3	143) 2	144) 2	145) 1
146) 2	147) 1	148) 3	149) 3	150) 1	151) 2	152) 4	153) 1	154) 2	155) 4
156) 2	157) 4	158) 4	159) 1	160) 2	161) 1	162) 1	163) 4	164) 2	165) 2
166) 1	167) 3	168) 2	169) 3	170) 1	171) 4	172) 3	173) 4	174) 1	175) 4
176) 1	177) 4	178) 3	179) 4	180) 3					

SOLUTIONS_PHYSICS _22-04-2020

91. Conceptual

92. $Q = nC_P d = n\frac{5}{2}RdT = 2\times\frac{5}{2}\times 8.3\times 5 = 208\,J$

93. $e = \frac{F\ell}{Ay} \Rightarrow \frac{e_1}{e_2} = \frac{\ell_1}{\ell_2}\times\frac{r_2^2}{r_1^2} = \frac{\ell}{2\ell}\times\frac{4r^2}{r^2} = 2$



94. $\frac{4K}{3}\frac{\Delta\theta\times 9}{2\ell} = \frac{3K\times 2A\theta t^1}{2\ell} \Rightarrow t_1 = 2\sec$

95. Magnetic field has no effect on the charged particle

$V = \sqrt{V_x^2 + V_y^2} \Rightarrow 4V_o^2 = V_o^2 + \left(\frac{Eq}{m}t\right)^2$

$\Rightarrow \frac{\sqrt{3}mV_o}{qE}$

96. $e = B\ell v = 20\times 10^{-4}$

$v = \frac{20\times 10^{-4}\times 4}{4}$

$= 20\times 10^{-4}$ volt

$\mathrm{I} = \frac{20\times 10^{-4}}{10} = 2\times 10^{-4} = 0.2\,\mathrm{mA}$

97. $P = \frac{1}{3}\rho V_{\text{rms}}^2$; $\rho = \frac{mn}{V} = \frac{mn}{\ell^3} \Rightarrow P = \frac{mnV^2}{3\ell^3}$

98. $\frac{GMm}{\ell}\times 2 = \frac{1}{2}mV^2$

$V = 2\sqrt{\frac{GM}{\ell}}$

99. $\left(\mathrm{v}_{\mathrm{He}}\right)_t = \frac{5}{7}\left(\mathrm{v}_{\mathrm{H_2}}\right)_{0°\mathrm{C}}$

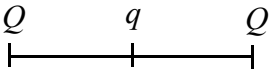
$\frac{273+t}{4} = \left(\frac{5}{7}\right)^2\frac{273}{2}$

$273 + t = \frac{25}{49}\times\frac{273}{2}\times 4 \simeq 273$

$\therefore t = 0°C$

100. Conceptual

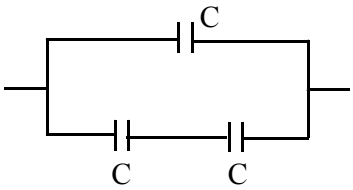
101. $K\left(\frac{QQ}{qx} + \frac{qQ}{x} + \frac{qQ}{x}\right) = 0$



$\frac{QQ}{2x} - 2\frac{qQ}{x} \Rightarrow q = -\frac{Q}{4}$

102. Equivalent circuit is

$C_{\text{eff}} = \frac{C}{2} + C = \frac{3}{2}C$



103. $\frac{X}{10} = \frac{52+1}{48+2}$

$X = \frac{53}{50}\times 10 = 10.6\,\Omega$

104. Total energy $= -\frac{GMm}{2(R+h)}$

Here $g_o = \frac{GM}{R^2}$ $\quad\therefore E = -\frac{mg_oR^2}{2(R+h)}$

105. The potential is represented by $V = 4x^2$ volt

$E = \frac{-dU}{dx} = -8x = -8\times 1 = 8\,\mathrm{Volt/m}$

106. Conceptual

107. $\eta = \frac{Q_2 - Q_1}{Q_1}$ and $P = \frac{1-\eta}{\eta} = \frac{1-1/10}{1/10} = 9$

$\beta = \frac{Q_2}{W} \Rightarrow Q_2 = \beta_W = 9\times 10 = 90\,J$

108. $\frac{1}{R}=\frac{1}{a}+\frac{1}{b}\Rightarrow R=\frac{ab}{a+b}$

109.

$$P_o+dg\frac{H}{2}+2dg\frac{H}{2}+0=P_o+2dgh+\frac{1}{2}\times 2d\times V^2$$

$$V=\sqrt{\frac{(3H-4h)g}{2}}$$

110. $BiAn=C_P\Rightarrow S_i=\frac{BAn}{C}$

$$\frac{S_1}{S_2}=\frac{9}{14}$$

111. $V_L=V_o\,;V_C=2V_o\,;V_R=3V_o$

$$V=\sqrt{V_R^2+(V_L-V_C)^2}=$$

$$\sqrt{9V_o^2+(V_o-2V_o)^2}=\sqrt{10V_o^2}$$

$10^4=10V_o^2\Rightarrow V_o=1000\Rightarrow V_o=10\sqrt{10}$

Now $V_R=3V_o=30\sqrt{10}\simeq 90$ volt

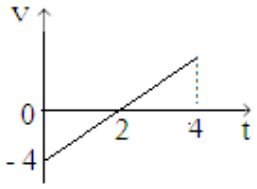
112. $0.2=\frac{2.1}{10+r}\Rightarrow 10+r=10.5\Rightarrow r=0.5\Omega$

113. $x=(t-2)^2, v=2(t-2)m/s$ and $a=2\ m/s^2$

Initial velocity (u) = -4 m/s

At t = 2s, v = 0

Final velocity at t = 4s is v = 4 m/s



Distance travelled

$=2\left[\frac{1}{2}\times 2\times 4\right]-4=8m$

114: Law of conservation of linear momentum is applied at maximum height of projectile.

$$m\times 60\bar{i}=\frac{m}{2}\times(50\bar{j})+\frac{m}{2}\bar{v}_2$$

$\Rightarrow \bar{v}_2=120\bar{i}-50\bar{j}$

115. $\mathrm{E}=\frac{\mathrm{v_B}}{\mathrm{d}}=\frac{0.3}{2\times 10^{-6}}=1.5\times 10^5\,\mathrm{v/m}$

Always from n to p side

116: For the sliding not to occur when

$$\tan\theta=\frac{dy}{dx}=\frac{2x}{a}=\mu\ \Rightarrow x=\frac{\mu a}{2}$$

$$y=\frac{x^2}{a}=\left(\frac{\mu a}{2}\right)^2\times\frac{1}{a}=\frac{1}{4}\mu^2 a$$

117. Emitter junction is forward bias and collector junction is in reverse bias then the transistor present in active region then it acts as an amplifies

118. Conceptual

119. Conceptual

120. $W=\frac{MgL}{2n^2}=\frac{MgL}{2(3)^2}=\frac{MgL}{18}$

(n = 3 given)

121: Torque of gravity

$\tau=mg\times x=mg\times(u\cos\theta)t$

122. From conservation of angular momentum

$$\frac{mr^2}{2}.w=mvr+\frac{mr^2}{2}\left(\frac{v}{r}\right)$$

$$\Rightarrow \mathrm{V}=\frac{rw}{3}$$

123: $V=\sqrt{\frac{2gh}{\left(1+\frac{k^2}{r^2}\right)}}=\sqrt{\frac{2gh}{(3/2)}}$

When it slides $V=\sqrt{2gH}$

$\sqrt{2gH}=\sqrt{\frac{4}{3}}gh; H=\frac{2h}{3}$

124. $F=bv$

$\Rightarrow (b)=\left(Kgs^{-1}\right)$

$y=Kt^2 \Rightarrow \frac{dy}{dt}=2kt$

125: $a=2k=2kms^{-2}$

$\frac{T_1}{T_2}=\sqrt{\frac{g=a}{g}}$

126. Observer hears two frequencies 1. n_1 which is coming from the source directly

2. n_2 which is coming from the reflection image of source

$n_1=680\left[\frac{340}{340-1}\right]$ and

$n_2=680\left[\frac{340}{340+1}\right] \Rightarrow n_1-n_2=4$

127. $y=\sin\pi x\cos 2\pi t$ Comparing with $y=2A\sin kx\cos\omega t;$ we have $\lambda=1m$. Minimum length of string first mode) $\left(L=\frac{\lambda}{2}\right)=0.5m$

128: In dispersion without deviation, mean colour (yellow) passes undeviated

129. Conceptual

130. Conceptual

131. Conceptual

132. As an electron makes a transition from an exited state to the ground state Kinetic energy increases but potential energy and total energy decrease

133. $K_{\max}=h\nu-h\nu_0 \quad \Rightarrow \quad eV_0=h\nu-h\nu_0 \Rightarrow$

$V_0=\frac{h}{e}\nu-\frac{h\nu_0}{e}$

Comparing this equation with $y=mx+c$, we get slope $m=\frac{h}{e} \Rightarrow h=m\times e$

134: $t_1=\frac{1}{\lambda}\log\frac{N_0}{N_1} \quad t_2=\frac{1}{\lambda}\log\frac{N_0}{N_2}$

$\frac{N_1}{N_0}=\frac{67}{100} \quad \frac{N_2}{N_0}=\frac{33}{100}$

$t_1=\frac{1}{\lambda}\log\frac{100}{67} \rightarrow (1) \; t_2=\frac{1}{\lambda}\log\frac{100}{33} \rightarrow (2)$

Solving (1) & (2) we get $t_2-t_1=20\min$

135. If k = 1 then the reactor is said to be in critical state

CHEMISTRY

136. 

No two electrons has same spin in an orbital

140. $\frac{r_{H_2}}{r_{O_2}}=\sqrt{\frac{32}{2}}=4$

141. 100 volume of H_2O_2 is perhydrol

142. Fe is ferro magnetic

MnO is antiferro magnetic

Fe_3O_4 is ferri magnetic

143. $\Delta T_f=K_f\times molarity\times i$

But $i = 1 + (2-1)0.3$

i = 1.3

$\therefore 0°C - x = 1.85 \times 0.2 \times 1.3$

$x = -0.481^0\text{C}$

144. $C_3H_6 + H_2 \rightarrow C_3H_8, \Delta H = -224kJ \rightarrow (4)$

$C_3H_8 + 5O_2 \rightarrow 3CO_3 + 4H_2O, \Delta H = -2027kJ \rightarrow (1)$

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O, \Delta H = -282kJ \rightarrow (2)$

$C_3H_6 + \frac{9}{2}O_2 \rightarrow 3CO_2 + 3H_2O, \Delta H = ? \rightarrow (3)$

$i.e.\ Eq.(4) = Eq.(3) + Eq.(2) - Eq.(1)$

-224 = x -282 + 2027

$\therefore x = -1969kJ$

145. $P^H = P^{ka} + \log\frac{N_sV_s}{N_{aV_a}}$

$= 4.8 + \log\frac{0.4 \times 100}{0.2 \times 20} = 4.8 + 1 = 5.8$

146. $K \propto \text{concentration} \times \text{number } of\ ions$

$\Lambda \text{ or } \mu \propto \frac{1}{\text{concentration} \times \text{number of ions}}$

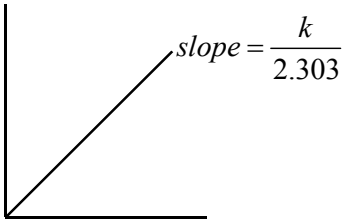
147. $\text{Q} = \text{c} \times \text{t} = 2 \times 965 = 0.02F$

i.e. $1F \rightarrow 1 mole\ of\ e^{-1}s$

0.02F $\rightarrow$? = 0.02 mole of $e^{-1}s$

149. $\log\frac{a}{(a-x)} = \frac{k}{2.303} \times t$

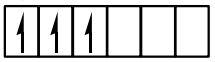
$y \qquad mx$



151. CIF_3 is sp^3d hybridized

157.

Cr^{+3} is $[Ar]4s^0 3d^3$

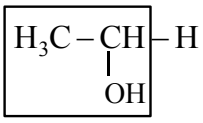


Unpaired electrons = 3

158. $\text{A} = H_3C - CH_2 - NH_2$

159. Ring – OH group has double bond character whereas side chain – OH group has single bond character which can be replaced easily by – chlorine

160. Ethyl alcohol has the required group



to answer haloform test

161. Peroxide has no effect on addition of HCl so X and Y are same compounds

162. In Friedal – Crafts alkylation, n – propyl carbonium ion under 1, 2 – hydride shift isopropyl carbonium ion as 2° carbonium ion is more stable to give isopropyl benzene (cumene)

164. They are Alkene derivates and single monomer

166. $22400\,\text{cm}^3 \rightarrow 6 \times 10^{23}$ molecules

$22.4\,\text{cm}^3 \rightarrow ?$

$= 6\times10^{20}$

167. $A^-_{(g)} \longrightarrow A_{(g)},\ \Delta H = +360\,KJ$

$A_{(g)} \longrightarrow A^-_{(g)},\ \Delta H = -360\,KJ$

168. Percentage of ionic character =

$$\frac{\mu_{observed}}{e\times d}\times 100$$

$$= \frac{2.4D}{4.8\times10^{-10}\times10^{8}\,esu.cm}\times 100$$

$$= \frac{2.4D}{4.8D}\times 100 = 50\%$$

169. $\Delta H = B.E_R - B.E_P$

$$-94 = \left(E_{N\equiv N} + 3E_{H-H}\right) - \left(6E_{N-H}\right)$$

$$-94 = \left(945 + 3\times436\right) - 6x$$

$$x = 391\,KJ$$

170. Geometrical due to $\left[Ma_4bc\right]^{+2}$ form

Linkage due to ambident ligand ONO

Hydrate due to inside and outside H_2O molecule

Ionisation due to producing Cl^- or Br^- ions

171. $AB_{(g)} \rightleftharpoons A_{(g)} + B_{(g)}$

1 0 0

$1-x$ x x where $x = 0.33 = \frac{1}{3}$

$$P_{AB} = P \times \frac{1-x}{1+x}$$

$$P_A = P \times \frac{x}{1+x}$$

$$P_B = P \times \frac{x}{1+x}$$

$$\therefore K_P = \frac{\left(P\times\frac{x}{1+x}\right)\left(P\times\frac{x}{1+x}\right)}{\left(P\times\frac{1-x}{1-x}\right)}$$

$$K_P = P\times\frac{x^2}{1-x^2} = P\times\frac{\left(\frac{1}{3}\right)^2}{1-\left(\frac{1}{3}\right)^2}$$

$$K_P = P\times\frac{1}{8}$$

$$\therefore P = 8K_P$$

172.

$$A = C_2H_5 - \overset{\overset{O}{\|}}{C} - CH_3$$

$$B = C_2H_5 - \underset{\underset{OH}{|}}{CH} - CH_3$$

$$C = C_2H_5 - \overset{\overset{O}{\|}}{C} - CH_3$$

'C' can't answer 'Tollens reagent'

173. $-O-CH_3$ group has more +I effect than $+CH_3$ group

175. $\Delta_o < P$ (weak acid)

$d^5 \longrightarrow t_{2g}^3 e_g^2$

$$\therefore CFSE = 3\left(-\frac{2}{5}\right) + 2\left(\frac{3}{5}\right)$$

CFSE = 0

177. Acetone hydrazone = $\left(\begin{matrix} H_3C \\ \diagdown \\ \quad C = N - NH_2 \\ \diagup \\ H_3C \end{matrix}\right)$

can't have geometrical isomerism

180. F^- is weak field ligand